

密级：



O2电层CLI手册v0.2

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武汉光迅科技股份有限公司

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密级：

文档修订记录

本文件拟制与解释部门：子系统开发部

版本	日期	主要起草部门专家	主要评审部门专家	修订情况
0.1	2022-01-01			初始版本

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密级:

1 登录到 O2 电层设备

用户可以通过串口或者网络 (ssh、telnet) 根据设备的 IP 地址登录到设备。设备出厂时, 默认用户名/密码 (admin/Admin 123)。

```
login as: admin
admin@10.3.29.10's password:

*****
*                                     *
*      / \ _/_/_/_/_/_/_/_/_/_/_\   *
*    _-_-/_/_/_/_/_/_/_/_/_/_\_./_  *
*   /_/_\/_/_/_/_/_/_/_/_/_/_\_/_  *
*  /_/_\/_/_/_/_/_/_/_/_/_/_\_/_  *
*                                     *
*****

<ACCELINK>
<ACCELINK> lis
<ACCELINK> list
cls
display alarm
display alarm all
display alarm notification
display alarm support-list
display current-configuration
display history
display ip-address
display loopback ip-address
display saved-configuration
display systime
display version
exit
list
quit
reboot
return
system-view
<ACCELINK>
```

登录到设备后，首先进入的是**基本视图**输入 `list` 命令可以获得全部命令。

密级：

2 基本视图命令

在基本视图中，可以执行一些简单的命令，了解系统的基本信息。

2.1 命令名称：cls

命令说明：

用来清除屏幕显示信息

```
undo sysname  
upload-log server (A.B.C.D) username STRING password STRING  
[AccelinkOLS]  
[AccelinkOLS]cls
```

```
[AccelinkOLS]
```

2.2 命令名称：list

命令说明：

用来显示当前视图下所有的命令

密级：

```
<AccelinkOLS> list
cls
display alarm
display alarm all
display alarm notification
display alarm support-list
display current-configuration
display history
display ip-address
display loopback ip-address
display saved-configuration
display systime
display version
exit
list
quit
reboot
return
system-view
<AccelinkOLS> sy
<AccelinkOLS> system-view
[AccelinkOLS]lis
[AccelinkOLS]list
clear slot <1-4> client (<1-12>|all) rmon (remote|local)
cls
cold reboot
config slot <1-4> line (<1-2>|all) rate (2|4|6|8|10|12) port-mode <1-6> bichm <0-127> fec-mode (0|1)
config slot <1-4> client (<1-12>|all) als (enable|disable)
config slot <1-4> client (<1-12>|all) als hold-off-time <50-2000>
config slot <1-4> client (<1-12>|all) lldp (enable|disable)
config slot <1-4> client (<1-12>|all) qsfp28_tx_laser_switch (enable|disable)
config slot <1-4> client (<1-12>|all) service (400ge|100ge|none)
config slot <1-4> line (<1-2>|all) freq STRING
display alarm
display alarm all
display alarm notification
display alarm support-list
display arp
display chassis manufacture
display cpu
display current-configuration
display eeprom mainboard
display fan
display ifconfig
display ifconfig all
display ip-address
display localtime
display log STRING <1-100>
```

2.3 命令名称：quit/exit

命令说明：

用来退出当前视图

```

<AccelinkOLS> ?
cls          Clear screen
display      Show alarm information
exit         Exit current mode and down to previous mode
list         Print command list
quit         Quit current mode and down to previous mode
reboot       Reboot the device
return       End current mode and change to enable mode.
system-view  Configuration terminal
<AccelinkOLS>
<AccelinkOLS> sy
<AccelinkOLS> system-view
[AccelinkOLS]
[AccelinkOLS]
[AccelinkOLS]?
clear        clear command
cls          Clear screen
cold         config cold reboot
config       config command
display      Show alarm information
exit         Exit current mode and down to previous mode
fan          Fan device
gateway      System default gateway address
get          get
hostname     Set hostname name
idle-timeout user time out
ip           IP information
list         Print command list
ntp          config ntp
ping         Network debug command
quit         Quit current mode and down to previous mode
reboot       Reboot the device
reset        Reset operation
return       End current mode and change to enable mode.
route        route protocol
save         Save current configuration
set          set
sftp         sftp
slot         The information of specify slot
system       Set system time
undo         Negate a command or set its defaults
upload-log   upload log to server command
[AccelinkOLS]
[AccelinkOLS]quit
<AccelinkOLS>
<AccelinkOLS> ?
cls          Clear screen
display      Show alarm information
exit         Exit current mode and down to previous mode
list         Print command list
quit         Quit current mode and down to previous mode
reboot       Reboot the device
return       End current mode and change to enable mode.

```

2.4 命令名称：return

命令说明：

用来退出当前视图

密级:

```
[AccelinkOLS]?
clear          clear command
cls           Clear screen
cold          config cold reboot
config        config command
display       Show alarm information
exit          Exit current mode and down to previous mode
fan           Fan device
gateway       System default gateway address
get           get
hostname      Set hostname name
idle-timeout  user time out
ip            IP information
list          Print command list
ntp           config ntp
ping          Network debug command
quit          Quit current mode and down to previous mode
reboot        Reboot the device
reset         Reset operation
return        End current mode and change to enable mode.
route         route protocol
save          Save current configuration
set           set
sftp          sftp
slot          The information of specify slot
systime       Set system time
undo          Negate a command or set its defaults
upload-log    upload log to server command
[AccelinkOLS]
[AccelinkOLS]
[AccelinkOLS]ret
[AccelinkOLS]return
<AccelinkOLS> ?
cls           Clear screen
display       Show alarm information
exit          Exit current mode and down to previous mode
list          Print command list
quit          Quit current mode and down to previous mode
reboot        Reboot the device
return        End current mode and change to enable mode.
system-view   Configuration terminal
<AccelinkOLS>
```

2.5 命令名称: reboot

命令说明:

reboot 命令用来软重启主控。

```
<AccelinkOLS> reboot
Confirm execute operation(y or n):
<AccelinkOLS>
<AccelinkOLS> syncing filesystems....This may take a few moments
Rebooting system!
The system is going down NOW!
Sending SIGTERM to all processes
Sending SIGKILL to all processes
Requesting system reboot

U-Boot 2010.12 (Dec 12 2017 - 18:23:33) V1.01

CPU: P1010, Version: 2.0, (0x80f10020)
Core: E500, Version: 5.2, (0x80212152)
Clock Configuration:
CPU0:800 MHz,
CCB:400 MHz,
DDR:400 MHz (800 MT/s data rate) (Asynchronous), IFC:100 MHz
L1: D-cache 32 kB enabled
I-cache 32 kB enabled
Board: P1010ACCE
I2C: ready
SPI: ready
DRAM: Configuring DDR for 667 MT/s data rate
DDR: 1 GiB (DDR3, 32-bit, CL=6, ECC off)
FLASH: 32 MiB
```

2.6 命令名称：display current-configuration

命令说明：

系统中，当前的一些设置信息，比如 IP 地址等，不通过 NETCONF 进行保存的，在这里进行显示。

```
[AccelinkOLS]
[AccelinkOLS]display current-configuration

Current Configuration:
sysname AccelinkOLS
ip eth1 address 10.3.29.169 mask 255.255.0.0
gateway eth1 add 10.3.29.254
ip eth2 address 192.168.100.200 mask 255.255.0.0
ntp disable
!

!end
[AccelinkOLS]
[AccelinkOLS]..
```

2.7 命令名称：display history

命令说明：

显示历史命令

```
<AccelinkOLS> ?
  cls          Clear screen
  display      Show alarm information
  exit         Exit current mode and down to previous mode
  list         Print command list
  quit         Quit current mode and down to previous mode
  reboot       Reboot the device
  return       End current mode and change to enable mode.
  system-view  Configuration terminal
<AccelinkOLS>
<AccelinkOLS>
<AccelinkOLS>
<AccelinkOLS> dis
<AccelinkOLS> display h
<AccelinkOLS> display history
  display version
  cls
  lis
  list
  system-view
  list
  cls
  q
  system-view
  quit
  q
  system-view
  quit
  return
  system-view
  return
  system-view
  display current-configuration
  display saved-configuration
<AccelinkOLS>
```


2.8 命令名称：display ip-address

命令说明：

显示 ip 地址信息

```
<AccelinkOLS> display ip-address
      Route 1 IP Addr      : 10.3.29.82
      Route 1 Mask        : 255.255.255.0
      Route 1 Gateway     : 10.3.29.254
      Route 1 Mac Addr    : 6c-4e-cc-22-22-93
      Route 2 IP Addr      :
      Route 2 Mask        :
      Route 2 Gateway     : 10.3.29.254
      Route 2 Mac Addr    : 6c-4e-cc-22-22-94

      lo:0 ip addr        :
      lo:0 mask           :

<AccelinkOLS>
```

2.9 命令名称：display systime

命令说明：

显示系统时间信息

```
<AccelinkOLS> display systime
2020-12-07 9:18:04
<AccelinkOLS>
```

2.10 命令名称：display version

命令说明：

显示系统版本信息

```
<AccelinkOLS> display version
Product Type      : OLS 02
System Running    : 0 Day 1 Hour 27 Minutes 4 Seconds
Startup Time      : 2021-11-04 17:32:48
Startup Type      : WARM REBOOT
Product Version   : V231R001C100B026 (Nov  4 2021 13:11:10)
Software Version  : 1.0.0
Hardware Version  : 3.0
ENM cpld version  : 19
Host Fpga Version : 0014
Panel Fpga Version: 0007

Device Unit      Serial Number      Product Number
CHASSIS          000A12148008      OLST-F-22
CU               000A02148008      OLST-EMU-22
LPU SLOT 1       000AAC15A981
LPU SLOT 2       000AAC518598
FAN GROUP 1      000A2015A701      OLST-FAN-22
FAN GROUP 2      000A2015B201      OLST-FAN-22
FAN GROUP 3      000A20148002      OLST-FAN-22
POWER SUPPLY 1   2K120065357      OLST-U-POWER-22
POWER SUPPLY 2   2K120065359      OLST-U-POWER-22

<AccelinkOLS>
```

2.11 命令名称：system-view

命令说明：

进入系统 view 窗口

```
<AccelinkOLS> ?
  cls          clear screen
  display      show alarm information
  exit         Exit current mode and down to previous mode
  list         Print command list
  quit         Quit current mode and down to previous mode
  reboot       Reboot the device
  return       End current mode and change to enable mode.
  system-view  Configuration terminal
<AccelinkOLS>
<AccelinkOLS> sy
<AccelinkOLS> system-view
[AccelinkOLS]
[AccelinkOLS]?
  clear      clear command
  cls        Clear screen
  cold       config cold reboot
  config     config command
  display    show alarm information
  exit       Exit current mode and down to previous mode
  fan        Fan device
  gateway    System default gateway address
  get        get
  hostname   Set hostname name
  idle-timeout user time out
  ip         IP information
  list       Print command list
  ntp        config ntp
  ping       Network debug command
  quit       Quit current mode and down to previous mode
  reboot     Reboot the device
  reset      Reset operation
  return     End current mode and change to enable mode.
  route      route protocol
  save       Save current configuration
  set        set
  sftp       sftp
  slot       The information of specify slot
  systime    Set system time
  undo       Negate a command or set its defaults
  upload-log upload log to server command
[AccelinkOLS]
```

3 系统视图相关命令行

在基本视图内执行 system-view 可以切换到系统视图，系统视图中可以对设备进行对应的管理。

3.1 系统基本属性

3.1.1 命令名称：sysname STRING

命令描述：

密级：

配置系统的 domain-name

```
[ACCELINK]
[ACCELINK] sy
[ACCELINK] sysname OPTEL
[OPTEL]
[OPTEL]
[OPTEL]
```

3.1.2 命令名称：timezone name WORD

命令描述：

配置系统的时区名称

```
[ACCELINK] timezone name
[ACCELINK] timezone name ?
WORD  UTC-11,UTC-10...UTC,UTC+1,UTC+2...UTC+12,Asia/Shanghai...
[ACCELINK] timezone name UTC-11
set timezone name success
[ACCELINK]
[ACCELINK]
[ACCELINK] display timezone
timezone name UTC_0
[ACCELINK]
[ACCELINK]
[ACCELINK] q
<ACCELINK>
<ACCELINK> dis
<ACCELINK> display sy
<ACCELINK> display systime
2020-12-06 16:54:39
<ACCELINK>
```

3.1.3 命令名称：display timezone STRING

命令描述：

查看系统当前的时区名称

```
[ACCELINK]
[ACCELINK] display timezone
timezone name UTC_19
[ACCELINK]
```

3.1.4 命令名称：sysptime <2000-2099>/<1-12>/<1-31>/<0-23>/<0-59>/<0-59>

命令描述：

配置当前的系统时间

密级：

```
<ACCELINK> display systime
2000-10-30 1:37:31
<ACCELINK>
<ACCELINK>
<ACCELINK> sy
<ACCELINK> system-view
[ACCELINK]
[ACCELINK] sy
[ACCELINK] systime 2020/12/7/10/37/55
2020/12/7, 10:37:55
[ACCELINK]
[ACCELINK] dis
[ACCELINK] q
<ACCELINK> dis
<ACCELINK> display sy
<ACCELINK> display systime
2020-12-07 10:38:04
<ACCELINK>
```

3.1.5 命令名称：display systime

命令描述：

显示当前的系统时间

```
[AccelinkOLS]
[AccelinkOLS]
[AccelinkOLS] display systime
2021-11-04 19:23:54, timezone:UTC+8
[AccelinkOLS]
[AccelinkOLS]
[AccelinkOLS]
```

3.1.6 命令名称：display version

命令说明：

显示系统版本信息

密级:

```
[ACCELINK]display version
Product Type       : OLS-O2
System Running     : 1 Day 8 Hours 16 Minutes 2 Seconds
Startup Time       : 2022-01-26 00:00:52
Startup Type       : WARM REBOOT
Product Version    : V231R001C100B035 (Jan  4 2022 16:56:03)
Software Version   : 1.0.0
Hardware Version   : 3.0
ENM Cpld Version   : 19
Host Fpga Version  : 0018
Panel Fpga Version : 0005

Device Unit        Serial Number      Product Number
CHASSIS            000A0115C708      O2-F
CU                 000A1115C708      O2-EMU
PANEL              000A1E15C708      O2-PANEL
FAN GROUP 1        000A2015C70D      O2-Fan
FAN GROUP 2        000A2015C70E      O2-Fan
FAN GROUP 3        000A2015C70F      O2-Fan
POWER SUPPLY 1     2K120052858      O2-PWR-B-2
POWER SUPPLY 2     2K120052857      O2-PWR-B-2
LPU SLOT 1         000AAC15C70A      T04X1-103
LPU SLOT 2         000AAC15C70B      T04X1-103
LPU SLOT 3         000AAC15C70C      T04X1-103
LPU SLOT 4         000AAC15C70E      T04X1-103
LPU SLOT 5         000AAC518598      T04X1-103
LPU SLOT 6         000AAC15C710      T04X1-103
LPU SLOT 7         000AAC15C711      T04X1-103
LPU SLOT 8         000AAC15C712      T04X1-103

[ACCELINK]
```

3.1.7 命令名称: save

命令描述:

保存在内存中的配置到对应的 CLI 文件中

```
[OPTEL(shadow)]
[OPTEL(shadow)]
[OPTEL(shadow)]shell/bin/sh: can't access tty: job control turned off
root@root:/mnt/application/www/www_ch#cat /mnt/flash/start-config.conf
!
! configuration saved from vty
!   2000/10/30 00:19:50
!
sysname OPTEL
!
root@root:/mnt/application/www/www_ch#exit

[OPTEL(shadow)]q
<OPTEL> sys
<OPTEL> system view
[OPTEL]sysname ACCELINK
[ACCELINK]save
Configuration saved to flash successful
[ACCELINK]
[ACCELINK]q
<ACCELINK> shadow node
Password:
[ACCELINK(shadow)]shell/bin/sh: can't access tty: job control turned off
root@root:/mnt/application/www/www_ch#cat /mnt/flash/start-config.conf
!
! configuration saved from vty
!   2020/12/07 11:48:33
!
sysname ACCELINK
ip eth1 address 172.16.100.200 mask 255.255.0.0
gateway eth1 add 172.16.1.1
ip eth2 address 192.168.100.200 mask 255.255.0.0
ntp disable
!
root@root:/mnt/application/www/www_ch#
root@root:/mnt/application/www/www_ch#
```

3.1.8 命令名称: idle-timeout <1-900>

命令描述:

配置命令行 CLI 的超时时间，在用户长时间没有输入的时候，会在配置的时间后自动退出当前的会话。

参数说明:

配置时间, 单位秒

```
[ACCELINK] idle-timeout
[ACCELINK] idle-timeout 10
[ACCELINK] vty connection is timed out.
[ACCELINK]
```

*
* A G G E L I N K *
*

Login:

3.2 网络相关

3.2.1 命令名称: ip (1 | 2) address (A.B.C.D) (A.B.C.D)

命令描述:

配置接口 NM1、NM2 的 ip 地址、子网掩码

```
[ACCELINK]
[ACCELINK]ip 1 address 10.3.29.178 255.255.255.0route: SIOCADDRT: File exists
route: SIOCADDRT: File exists
```

```
set ip 1 address 10.3.29.178 255.255.255.0 success.
<success>
set ip 1 gateway 10.3.29.254 success.
[ACCELINK]
[ACCELINK]
[ACCELINK]ip 2 address 192.168.100.200 255.255.255.0
<success>
set ip 2 address 192.168.100.200 255.255.255.0 success.
<fail>
set ip 2 gateway fail.
[ACCELINK]
[ACCELINK]display ip-address
```

```
Route 1 IP Addr      : 10.3.29.178
Route 1 Mask         : 255.255.255.0
Route 1 Gateway      : 10.3.29.254
Route 1 Mac Addr     : 6c-4e-cc-72-22-ab
Route 2 IP Addr      : 192.168.100.200
Route 2 Mask         : 255.255.255.0
Route 2 Gateway      :
Route 2 Mac Addr     : 6c-4e-cc-72-22-ac

lo:0 ip addr         :
lo:0 mask             :
```

[ACCELINK]

3.2.2 命令名称：ip (1 | 2) address del

命令描述：

删除接口 NM1、NM2 的 ip 地址、子网掩码

```
[ACCELINK]ip 1 address del route: SIOCADDRT: No such process
<success>
set ip 1 address del success.
[ACCELINK]
[ACCELINK]
[ACCELINK]ip 2 address del
<success>
set ip 2 address del success.
[ACCELINK]
```

3.2.3 命令名称：ip loopback address (A.B.C.D) mask (A.B.C.D)

命令描述：

配置 loopback 的 ip 地址

```
[ACCELINK]display loopback ip-address
The device don't config loopback ip information!!
[ACCELINK]
[ACCELINK]ip
[ACCELINK]ip
eth1      eth2      loopback
[ACCELINK]ip loo
[ACCELINK]ip loopback
[ACCELINK]ip loopback address
[ACCELINK]ip loopback address 1.1.1.1
[ACCELINK]ip loopback address 1.1.1.1 mask 255.255.255.255
[ACCELINK]
[ACCELINK]
[ACCELINK]display loopback ip-address

Loopback ip information:
lo:0 ip addr      : 1.1.1.1
lo:0 mask         : 255.255.255.255

[ACCELINK]
[ACCELINK]ip
[ACCELINK]ip
eth1      eth2      loopback
[ACCELINK]ip loo
[ACCELINK]ip loopback
[ACCELINK]ip loopback address
[ACCELINK]ip loopback address del
[ACCELINK]
[ACCELINK]display loopback ip-address

Loopback ip information:
lo:0 ip addr      :
lo:0 mask         :

[ACCELINK]
```

3.2.4 命令名称：ip loopback address del

命令描述：

删除 loopback 的 ip 地址

密级：

```
[ACCELINK]display loopback ip-address
The device don't config loopback ip information!!
[ACCELINK]
[ACCELINK]ip
[ACCELINK]ip
eth1      eth2      loopback
[ACCELINK]ip loo
[ACCELINK]ip loopback
[ACCELINK]ip loopback address
[ACCELINK]ip loopback address 1.1.1.1
[ACCELINK]ip loopback address 1.1.1.1 mask 255.255.255.255
[ACCELINK]
[ACCELINK]
[ACCELINK]display loopback ip-address

Loopback ip information:
lo:0 ip addr      : 1.1.1.1
lo:0 mask         : 255.255.255.255

[ACCELINK]
[ACCELINK]ip
[ACCELINK]ip
eth1      eth2      loopback
[ACCELINK]ip loo
[ACCELINK]ip loopback
[ACCELINK]ip loopback address
[ACCELINK]ip loopback address del
[ACCELINK]
[ACCELINK]display loopback ip-address

Loopback ip information:
lo:0 ip addr      :
lo:0 mask         :

[ACCELINK]
```

3.2.5 命令名称：ping (A.B.C.D)

命令描述：

Ping 命令，用来网络诊断

```
[ACCELINK]
[ACCELINK]ping 10.3.22.143
PING 10.3.22.143 (10.3.22.143): 56 data bytes
64 bytes from 10.3.22.143: seq=0 ttl=128 time=2.369 ms
64 bytes from 10.3.22.143: seq=1 ttl=128 time=0.724 ms
64 bytes from 10.3.22.143: seq=2 ttl=128 time=0.774 ms
64 bytes from 10.3.22.143: seq=3 ttl=128 time=100.014 ms

--- 10.3.22.143 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.724/25.970/100.014 ms

[ACCELINK]
```

3.2.6 命令名称：ping (A.B.C.D) <1-50>

命令描述：

Ping 命令，用来网络诊断，指定 ping 包个数

密级：

```
[AccelinkOLS]ping 10.3.56.143 2
PING 10.3.56.143 (10.3.56.143): 56 data bytes
64 bytes from 10.3.56.143: seq=0 ttl=128 time=2.292 ms
64 bytes from 10.3.56.143: seq=1 ttl=128 time=0.494 ms

--- 10.3.56.143 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.494/1.393/2.292 ms

[AccelinkOLS]
```

3.2.7 命令名称：ping (A.B.C.D) source (A.B.C.D)

命令描述：

Ping 命令，用来网络诊断，指定源 ip

```
[ACCELINK]ping 10.3.22.143 sou
[ACCELINK]ping 10.3.22.143 source 10.3.22.68
PING 10.3.22.143 (10.3.22.143) from 10.3.22.68: 56 data bytes
64 bytes from 10.3.22.143: seq=0 ttl=128 time=0.640 ms
64 bytes from 10.3.22.143: seq=1 ttl=128 time=0.614 ms
64 bytes from 10.3.22.143: seq=2 ttl=128 time=0.480 ms
64 bytes from 10.3.22.143: seq=3 ttl=128 time=0.645 ms

--- 10.3.22.143 ping statistics ---
4 packets transmitted, 4 packets received, 0% packet loss
round-trip min/avg/max = 0.480/0.594/0.645 ms

[ACCELINK]
```

3.2.8 命令名称：route (zebra|ospfd|bgpd)

命令描述：

网络路由命令

参数说明：

选择进入 zebra 或者 ospf, bgp 的控制台

密级：

```
[ACCELINK]route
bgpd      ospfd      zebra
[ACCELINK]route z
[ACCELINK]route zebra
Router>
Router>
Router>
  enable      Turn on privileged mode command
  exit        Exit current mode and down to previous mode
  find        Find CLI command matching a regular expression
  help        Description of the interactive help system
  list        Print command list
  quit        Exit current mode and down to previous mode
  show        Show running system information
  terminal    Set terminal line parameters
  who         Display who is on vty
Router>
Router>
Router> q

[ACCELINK]route bgp
[ACCELINK]route bgpd
bgpd>
  enable      Turn on privileged mode command
  exit        Exit current mode and down to previous mode
  find        Find CLI command matching a regular expression
  help        Description of the interactive help system
  list        Print command list
  quit        Exit current mode and down to previous mode
  show        Show running system information
  terminal    Set terminal line parameters
  who         Display who is on vty
bgpd> q

[ACCELINK]route z
[ACCELINK]route zebra
Router>
  enable      Turn on privileged mode command
  exit        Exit current mode and down to previous mode
  find        Find CLI command matching a regular expression
  help        Description of the interactive help system
  list        Print command list
  quit        Exit current mode and down to previous mode
  show        Show running system information
  terminal    Set terminal line parameters
  who         Display who is on vty
Router>
Router>
Router> q

[ACCELINK]
```

3.2.9 命令名称：gateway (1 | 2) add (A.B.C.D)

命令描述：

给端口添加网关地址

密级：

```
[ACCELINK]gateway 1 add 10.3.129.254
[ACCELINK]gateway 1 add 10.3.129.254route: SIOCADDRT: File exists

<success>
set gateway 1 add 10.3.129.254 success.
[ACCELINK]di
[ACCELINK]display ip
[ACCELINK]display ip-address

Route 1 IP Addr      : 10.3.129.100
Route 1 Mask         : 255.255.255.0
Route 1 Gateway      : 10.3.129.254
Route 1 Mac Addr     : 6c-2e-33-06-ff-bb
Route 2 IP Addr      : 192.168.100.200
Route 2 Mask         : 255.255.0.0
Route 2 Gateway      :
Route 2 Mac Addr     : 6c-2e-33-06-ff-bc

lo:0 ip addr         :
lo:0 mask             :
```

3.2.10 命令名称：gateway (1 | 2) del

命令描述：

给端口删除网关配置

参数说明：

指定对应的端口，eth1 或者 eth2

```
[ACCELINK]gateway 1 del
[ACCELINK]gateway 1 del
<success>
set gateway 1 del success.
[ACCELINK]di
[ACCELINK]display ip
[ACCELINK]display ip-address

Route 1 IP Addr      : 10.3.129.100
Route 1 Mask         : 255.255.255.0
Route 1 Gateway      :
Route 1 Mac Addr     : 6c-2e-33-06-ff-bb
Route 2 IP Addr      : 192.168.100.200
Route 2 Mask         : 255.255.0.0
Route 2 Gateway      :
Route 2 Mac Addr     : 6c-2e-33-06-ff-bc

lo:0 ip addr         :
lo:0 mask             :

[ACCELINK]
```

3.2.11 命令名称：display ip-address

命令描述：

显示当前设备的 ip 地址信息

密级：

```
<AccelinkOLS> display ip-address

Eth ip information:
  Route 1 ip addr      : 10.3.29.169
  Route 1 mask         : 255.255.0.0
  Route 1 gateway      : 10.3.29.254
  Route 1 mac addr     : 6c-2e-33-00-d0-72
  Route 2 ip addr      : 192.168.100.200
  Route 2 mask         : 255.255.0.0
  Route 2 mac addr     : 6c-2e-33-00-d0-73

<AccelinkOLS>
```

3.2.12 命令名称：display tcp session

命令描述：

用来查询设备 tcp 会话

```
[ACCELINK]display tcp session
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp      0      0 10.3.29.169:ssh        10.3.56.189:51254      ESTABLISHED
tcp      0      0 ::ffff:10.3.29.169%3218794544:830 ::ffff:10.3.56.189:65343 ESTABLISHED
[ACCELINK]
[ACCELINK]
```

3.2.13 命令名称：display route

命令描述：

用来查询设备路由信息

```
[AccelinkOLS]display route
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
117.103.88.0 * 255.255.248.0 U 0 0 0 eth2
10.3.0.0 * 255.255.0.0 U 0 0 0 eth1.1
default 10.3.29.254 0.0.0.0 UG 0 0 0 eth1.1
[AccelinkOLS]
```

3.2.14 命令名称：display process

命令描述：

用来查询设备进程

密级：

```
[AccelinkOLS]display process
PID USER      VSZ STAT COMMAND
  1 root        4200 S    init
  2 root         0 SW    [kthreadd]
  3 root         0 SW    [ksoftirqd/0]
  4 root         0 SW    [watchdog/0]
  5 root         0 SW    [events/0]
  6 root         0 SW    [khelper]
  9 root         0 SW    [async/mgr]
153 root         0 SW    [sync_supers]
155 root         0 SW    [bdi-default]
157 root         0 SW    [kblockd/0]
163 root         0 SW    [ata_aux]
164 root         0 SW    [ata_sff/0]
172 root         0 SW    [khubd]
175 root         0 SW    [kseriod]
192 root         0 SW    [rpciod/0]
248 root         0 SW    [khungtaskd]
249 root         0 SW    [kswapd0]
302 root         0 SW    [aio/0]
314 root         0 SW    [nfsiod]
322 root         0 SW    [crypto/0]
1035 root         0 SW    [mtdblock0]
1040 root         0 SW    [mtdblock1]
1045 root         0 SW    [mtdblock2]
1050 root         0 SW    [mtdblock3]
1055 root         0 SW    [mtdblock4]
1060 root         0 SW    [mtdblock5]
1065 root         0 SW    [mtdblock6]
1073 root         0 SW    [mtdblock7]
1078 root         0 SW    [mtdblock8]
1083 root         0 SW    [mtdblock9]
1088 root         0 SW    [mtdblock10]
1093 root         0 SW    [mtdblock11]
1098 root         0 SW    [mtdblock12]
1103 root         0 SW    [mtdblock13]
1108 root         0 SW    [mtdblock14]
1113 root         0 SW    [mtdblock15]
1180 root         0 SW    [edac-poller]
1198 root         0 SW    [kjournald]
1256 root        2896 S    /usr/sbin/vsftpd
1258 root        3420 S    /usr/sbin/time_update -d -t 60
1259 root        114m S    ./test
1268 root        5164 S    /usr/sbin/sshd
1277 root         0 SW    [ubi_bgt0d]
1280 root         0 SW    [ubifs_bgt0_0]
1294 root        4200 S    /bin/sh /etc/coreFilewatch.sh
1320 root       53280 S    ./zebra -d -g root -u root -f /mnt/flash/zebra.conf
1324 root      39236 S    ./bgpd -d -g root -u root -f /mnt/flash/bgpd.conf
```

3.2.15 命令名称：display netstat all

命令描述：

用来查询设备 netstat 信息

密级:

```
[ACCELINK]display netstat all
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:zebra           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ospfd           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:bgpd            0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ftp             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:telnet           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:https            0.0.0.0:*               LISTEN
tcp        0      0 :::%3215526144:zebra   :::*                   LISTEN
tcp        0      0 :::%3215526144:ospfd   :::*                   LISTEN
tcp        0      0 :::%3215526144:bgpd    :::*                   LISTEN
tcp        0      0 :::%3215526144:ssh     :::*                   LISTEN
tcp        0      0 :::%3215526144:830     :::*                   LISTEN
udp        0      0 0.0.0.0:9901            0.0.0.0:*               *
udp        0      0 0.0.0.0:9000            0.0.0.0:*               *
udp        0      0 localhost:8002          0.0.0.0:*               *
raw        0      0 0.0.0.0:89             0.0.0.0:*               89
raw        0      0 :::%3215526144:58     :::*                   58
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags               Type               State              I-Node Path
unix    2      [ ACC ]              STREAM             LISTENING          1281 /var/run/zserv.api
unix    2      [ ACC ]              STREAM             LISTENING          1287 /var/run/zebra.vty
unix    2      [ ACC ]              STREAM             LISTENING          1317 /var/run/bgpd.vty
unix    2      [ ACC ]              STREAM             LISTENING          1345 /var/run/ospfd.vty
unix    2      [ ACC ]              STREAM             LISTENING          2565 /var/run/lldpd.socket
unix    5                        DGRAM              1968 /dev/log
unix    2                        DGRAM              694835
unix    3      [ ]              STREAM             CONNECTED          2575
unix    3      [ ]              STREAM             CONNECTED          2574
unix    3      [ ]              STREAM             CONNECTED          2570
unix    3      [ ]              STREAM             CONNECTED          2569
unix    2      [ ]              DGRAM              2560
unix    2      [ ]              DGRAM              2237
unix    3      [ ]              STREAM             CONNECTED          1348 /var/run/zserv.api
unix    3      [ ]              STREAM             CONNECTED          1347
unix    3      [ ]              STREAM             CONNECTED          1323 /var/run/zserv.api
unix    3      [ ]              STREAM             CONNECTED          1322
unix    3      [ ]              STREAM             CONNECTED          1320 /var/run/zserv.api
unix    3      [ ]              STREAM             CONNECTED          1319
[ACCELINK] ..
```

3.2.16 命令名称: display ifconfig (all)

命令描述:

用来查询设备系统 linux 的 ifconfig 信息

```
[AccelinkOLS]display ifconfig
eth1      Link encap:Ethernet  Hwaddr 6C:2E:33:00:00:02
          UP BROADCAST RUNNING PROMISC MULTICAST  MTU:2000  Metric:1
          RX packets:1754633 errors:0 dropped:0 overruns:0 frame:0
          TX packets:956043 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:146362544 (139.5 MiB)  TX bytes:1215905053 (1.1 GiB)
          Base address:0xe000

eth1.1    Link encap:Ethernet  Hwaddr 6C:2E:33:00:D0:72
          inet addr:10.3.29.169  Bcast:10.3.255.255  Mask:255.255.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:2000  Metric:1
          RX packets:1754521 errors:0 dropped:0 overruns:0 frame:0
          TX packets:956042 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:93704690 (89.3 MiB)  TX bytes:1215905011 (1.1 GiB)

eth1.3    Link encap:Ethernet  Hwaddr 6C:2E:33:01:D1:74
          UP BROADCAST RUNNING MULTICAST  MTU:2000  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

```

[AccelinkOLS]display ifconfig all
eth0      Link encap:Ethernet  Hwaddr 00:04:9F:EF:01:02
          BROADCAST MULTICAST  MTU:1500  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
          Base address:0x6000

eth1      Link encap:Ethernet  Hwaddr 6C:2E:33:00:00:02
          UP BROADCAST RUNNING PROMISC MULTICAST  MTU:2000  Metric:1
          RX packets:1755483 errors:0 dropped:0 overruns:0 frame:0
          TX packets:956532 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:146434243 (139.6 MiB)  TX bytes:1216544616 (1.1 GiB)
          Base address:0xe000

eth1.1    Link encap:Ethernet  Hwaddr 6C:2E:33:00:D0:72
          inet addr:10.3.29.169  Bcast:10.3.255.255  Mask:255.255.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:2000  Metric:1
          RX packets:1755371 errors:0 dropped:0 overruns:0 frame:0
          TX packets:956531 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:93750889 (89.4 MiB)  TX bytes:1216544574 (1.1 GiB)

eth1.2    Link encap:Ethernet  Hwaddr 6C:2E:33:00:D0:73
          inet addr:192.168.100.200  Bcast:192.168.255.255  Mask:255.255.0.0
          BROADCAST MULTICAST  MTU:2000  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:1 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:42 (42.0 B)

eth1.3    Link encap:Ethernet  Hwaddr 6C:2E:33:01:D1:74
          UP BROADCAST RUNNING MULTICAST  MTU:2000  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

```

3.3 NTP 模块

3.3.1 命令名称：ntp (enable|disable)

命令描述：

打开设备的 NTP 功能

```

[ACCELINK]display ntp
[ACCELINK]display ntp
ntp not enabled yet!
[ACCELINK]
[ACCELINK]ntp
[ACCELINK]ntp
disable enable server
[ACCELINK]ntp en
[ACCELINK]ntp enable
[ACCELINK]ntp enable
[ACCELINK]ntp enable
ntp enable successful.
[ACCELINK]
[ACCELINK]display ntp
ntp server ip : 10.3.56.27
[ACCELINK]
[ACCELINK]

```


3.3.2 命令名称：ntp server add (A.B.C.D)

命令描述：

增加 NTP 服务区的 server 地址

```
[ACCELINK]display ntp
ntp server not config yet.
[ACCELINK]ntp
[ACCELINK]ntp se
[ACCELINK]ntp server
add del
[ACCELINK]ntp server
add del
[ACCELINK]ntp server ad
[ACCELINK]ntp server add
[ACCELINK]ntp server add 10.3.22.143
ntp server add 10.3.22.143 successful.
[ACCELINK]
[ACCELINK]ntp server add 10.3.22.144
ntp server add 10.3.22.144 successful.
[ACCELINK]
[ACCELINK]display ntp
ntp server ip          : 10.3.22.144
                        : 10.3.22.143
[ACCELINK]
```

3.3.3 命令名称：ntp server del (A.B.C.D)

命令描述：

删除一个 NTP 服务器地址

```
[ACCELINK]display ntp
ntp server ip          : 10.3.56.27
[ACCELINK]
[ACCELINK]net
[ACCELINK]ntp
[ACCELINK]ntp se
[ACCELINK]ntp server
add del
[ACCELINK]ntp server
add del
[ACCELINK]ntp server d
[ACCELINK]ntp server del
[ACCELINK]ntp server del
% Command incomplete.
[ACCELINK]ntp server del 10.3.56.27
ntp server del 10.3.56.27 successful.
[ACCELINK]
[ACCELINK]display ntp
ntp server not config yet.
[ACCELINK]
```

3.3.4 命令名称：display ntp

命令描述：

显示当前 NTP 的状态

密级：

```
[ACCELINK] ntp
[ACCELINK] display nt
[ACCELINK] display ntp
ntp not enabled yet!
[ACCELINK]
[ACCELINK] ntp
[ACCELINK] ntp
disable enable server
[ACCELINK] ntp en
[ACCELINK] ntp enable
[ACCELINK] ntp enable
[ACCELINK] ntp enable
ntp enable successful.
[ACCELINK]
[ACCELINK] display ntp
ntp server ip : 10.3.56.27
[ACCELINK]
[ACCELINK]
```

3.4 设备管理

3.4.1 命令名称：reboot

命令描述：

软重启主控 CPU

```
<AccelinkOLS> reboot
Confirm execute operation(y or n):
<AccelinkOLS>
<AccelinkOLS> syncing filesystems....This may take a few moments
Rebooting system!
The system is going down NOW!
Sending SIGTERM to all processes
Sending SIGKILL to all processes
Requesting system reboot

U-Boot 2010.12 (Dec 12 2017 - 18:23:33) v1.01

CPU: P1010, Version: 2.0, (0x80f10020)
Core: E500, Version: 5.2, (0x80212152)
Clock configuration:
CPU0:800 MHz,
CCB:400 MHz,
DDR:400 MHz (800 MT/s data rate) (Asynchronous), IFC:100 MHz
L1: D-cache 32 kB enabled
I-cache 32 kB enabled
Board: P1010ACCE
I2C: ready
SPI: ready
DRAM: Configuring DDR for 667 MT/s data rate
DDR: 1 GiB (DDR3, 32-bit, CL=6, ECC off)
FLASH: 32 MiB
```

3.4.2 命令名称：cold reboot

命令描述：

冷重启主控 CPU

密级：

```
[Accelink]cold reboot
Confirm execute operation(y or n):
U-Boot 2010.12 (Dec 12 2017 - 18:23:33) v1.01

CPU: P1010, Version: 2.0, (0x80f10020)
Core: E500, Version: 5.2, (0x80212152)
Clock Configuration:
  CPU0:800 MHz,
  CCB:400 MHz,
  DDR:400 MHz (800 MT/s data rate) (Asynchronous), IFC:100 MHz
L1: D-cache 32 kB enabled
  I-cache 32 kB enabled
Board: P1010ACCE
I2C: ready
SPI: ready
DRAM: Configuring DDR for 667 MT/s data rate
DDR: 1 GiB (DDR3, 32-bit, CL=6, ECC off)
FLASH: 32 MiB
L2: 256 kB enabled
NAND: NAND device: Manufacturer ID: 0x01, Chip ID: 0xd3 (AMD NAND 1GiB 3,3V 8-bit)
1024 MiB
PCIe1: Root Complex of mini PCIe slot, no link, regs @ 0xffe0a000
PCIe1: Bus 00 - 00
PCIe2: Root Complex of PCIe slot, no link, regs @ 0xffe09000
PCIe2: Bus 01 - 01
In: serial
Out: serial
Err: serial
Net: eTSEC1: Not support
  eTSEC2: No support for PHY id ffffffff; assuming generic
  eTSEC3: No support for PHY id ffffffff; assuming generic
  eTSEC1, eTSEC2, eTSEC3
CPLD : v1.5
Hit ctrl+b to stop autoboot: 1 █
```

3.4.3 命令名称：display cpu

命令描述：

显示主控上 CPU 的当前状态

```
[ACCELINK]display cpu
Used(%)      Threshold(%)      Status
15           80           Normal
[ACCELINK]
```

3.4.4 命令名称：display memory

命令描述：

显示主控内存的当前状态

```
[ACCELINK]display memory
Total(KB)      Free(KB)      Used(%)      Threshold(%)      Status
1032660        813620        21.2         80           Normal
[ACCELINK]
```

3.4.5 命令名称：display temperature

命令描述：

显示当前温度信息

密级：

```
[AccelinkOLS]display temperature
Temperature information:
  CU-1-41 temperature       : cur: 39.50 C  peak: 39.50 C
  LINECARD-1-1 temperature : 54.8 C
  LINECARD-1-2 temperature : 74.3 C
  LINECARD-1-3 temperature : 67.4 C
  LINECARD-1-4 temperature : 67.3 C
  FAN-1-31 temperature      : 39.1 C
  FAN-1-32 temperature      : 36.5 C
  FAN-1-33 temperature      : 34.8 C
[AccelinkOLS]
```

3.5 风扇电源管理

3.5.1 命令名称：display fan

命令描述：

显示当前风扇的状态

```
[AccelinkOLS]display fan
There are 3 fan online:
  No. SN           Status    Mode    Lv RPM_A  RPM_B
  --- --           -
  1  000A2011A205  Normal   Auto    1  3963  3735
  2  000A2014BC81  Normal   Auto    1  3957  3717
  3  000A2015A083  Normal   Auto    1  4010  3764
[AccelinkOLS]
```

3.5.2 命令名称：fan id (1|2|3|all) speed-level <1-5>

命令描述：

手动控制风扇的转速(系统默认为 auto 模式)

参数说明：

指定风扇的编号，或者全部

选择风扇的挡位

密级：

```
[AccelinkOLS]display fan
There are 3 fan online:
  No. SN           Status    Mode    Lv RPM_A  RPM_B
  1   000A2011A205 Normal    Auto    1  4418  4189
  2   000A2014BC81 Normal    Auto    1  2961  2754
  3   000A2015A083 Normal    Auto    1  2865  2775

[AccelinkOLS]
[AccelinkOLS]fan id 1 speed-level 2
<success>
set fan id 1 speed-level 2 success.
[AccelinkOLS]
[AccelinkOLS]display fan
There are 3 fan online:
  No. SN           Status    Mode    Lv RPM_A  RPM_B
  1   000A2011A205 Normal    Manual  2  4424  4189
  2   000A2014BC81 Normal    Auto    2  3663  3452
  3   000A2015A083 Normal    Auto    2  3722  3480

[AccelinkOLS]
```

3.5.3 命令名称：fan mode auto

命令描述：

设置控制风扇自动调速(系统默认为 auto 模式)

参数说明：

```
[AccelinkOLS]fan mode auto
<success>
set fan mode auto success.
[AccelinkOLS]
[AccelinkOLS]display fan
There are 3 fan online:
  No. SN           Status    Mode    Lv RPM_A  RPM_B
  1   000A2011A205 Normal    Auto    2  4418  4184
  2   000A2014BC81 Normal    Auto    2  3488  3239
  3   000A2015A083 Normal    Auto    2  3444  3184

[AccelinkOLS]
```

3.5.4 命令名称：fan <1-3> sn <2000-2099>/<1-12>/<1-31> number <0-127>

命令描述：

设置风扇 SN(shadow 视图下配置)。风扇 SN 设置规则为，延续 OSP 协议中的 6 字节 SN 规则：保留 8bit+保留 4bit+产品大类 4bit+产品小类 8bit+年 8bit+月 4bit+日 5bit+流水号 7bit。

参数说明：

```
[AccelinkOLS(shadow)]fan 1 sn 2021/11/4 number 3
set fan 1 SN 000a2015b203 success !
[AccelinkOLS(shadow)]
```


3.5.5 命令名称：display power

命令描述：

显示电源的状态

```
[AccelinkOLS]display power
There are 2 power supply online:
Power Supply No.1 information:
Power Input      : 0.00 V, 0.00 A
Power Output     : 0.00 V, 0.00 A
Pin Pout PoutMax : 0.00 W, 0.00 W, 800.00 W
Power Temperature 1 : 28.31 C
Power Temperature 2 : 27.56 C
Power Temperature 3 : 38.12 C
Power PN         : GW-CRPS800N2
Power SN         : 2K120065357

Power Supply No.2 information:
Power Input      : 223.75 V, 0.78 A
Power Output     : 12.23 V, 13.14 A
Pin Pout PoutMax : 174.53 W, 161.00 W, 800.00 W
Power Temperature 1 : 31.12 C
Power Temperature 2 : 40.50 C
Power Temperature 3 : 39.93 C
Power PN         : GW-CRPS800N2
Power SN         : 2K120065359

[AccelinkOLS]
```

3.6 LOG 模块

3.6.1 命令名称：display log STRING <1-100>

命令描述：

显示指定的 log 文件中的部分内容

参数说明：

指定的 log 文件名称，以及对应的行号

先用 display log file 来查看具体的 log 文件名称

```
[AccelinkOLS]
[AccelinkOLS]display log error.1417.log 10
2020-11-19 13:33:33 AccelinkOLS/INFO [185-56]:itemId:9701, portId:2, portType:1
2020-11-19 13:33:33 AccelinkOLS/INFO [185-57]:itemId:6801, portId:2, portType:1
2020-11-19 13:33:33 AccelinkOLS/INFO [185-58]:itemId:6901, portId:2, portType:1
2020-11-19 13:33:33 AccelinkOLS/INFO [185-59]:itemId:6a01, portId:2, portType:1
2020-11-19 13:33:33 AccelinkOLS/INFO [185-60]:itemId:4701, portId:2, portType:1
2020-11-19 13:33:33 AccelinkOLS/INFO [185-61]:itemId:4501, portId:2, portType:1
2020-11-19 13:33:33 AccelinkOLS/INFO [185-62]:itemId:4901, portId:2, portType:1
2020-11-19 13:33:33 AccelinkOLS/INFO [185-63]:itemId:4a01, portId:2, portType:1
2020-11-19 13:33:34 AccelinkOLS/INFO [185-64]:itemId:4b01, portId:2, portType:1
2020-11-19 13:33:34 AccelinkOLS/INFO [185-65]:itemId:4c01, portId:2, portType:1
[AccelinkOLS]
[AccelinkOLS]
```

3.6.2 命令名称：display log file

命令描述：

密码:

列出当前系统中存在的 log 文件列表

```
[AccelinkOLS]display log file
-rw----- 1 root root 27480 Dec 4 17:22 alarm.log
-rw----- 1 root root 7545 Nov 27 16:45 debug.log
-rw----- 1 root root 999942 Nov 19 13:33 error.1417.log
-rw----- 1 root root 999968 Nov 19 13:36 error.1418.log
-rw----- 1 root root 999900 Nov 19 13:39 error.1419.log
-rw----- 1 root root 999895 Nov 19 13:42 error.1420.log
-rw----- 1 root root 999961 Nov 19 13:45 error.1421.log
-rw----- 1 root root 999966 Nov 19 13:48 error.1422.log
-rw----- 1 root root 257025 Dec 5 22:51 error.log
-rw----- 1 root root 999988 Oct 19 11:33 event.0.log
-rw----- 1 root root 1000003 Nov 3 07:45 event.1.log
-rw----- 1 root root 999749 Nov 16 16:09 event.2.log
-rw----- 1 root root 999900 Dec 5 17:03 event.3.log
-rw----- 1 root root 14747 Dec 7 14:33 event.log
-rw-rw-rw- 1 root root 30752 Sep 21 10:02 message
-rw-rw-rw- 1 root root 204843 Sep 21 10:02 message.0
-rw----- 1 root root 0 Sep 2 16:03 otdr.log
-rw----- 1 root root 11195 Dec 4 16:44 reboot.log
-rw----- 1 root root 213400 Sep 21 10:02 trap.log
-rw----- 1 root root 71666 Nov 27 16:45 upgrade.log
-rw----- 1 root root 999940 Sep 9 15:46 user.0.log
-rw----- 1 root root 999984 Sep 12 11:55 user.1.log
-rw----- 1 root root 999872 Nov 16 09:26 user.10.log
-rw----- 1 root root 999973 Sep 14 15:37 user.2.log
-rw----- 1 root root 999949 Sep 16 09:53 user.3.log
-rw----- 1 root root 999995 Sep 16 16:15 user.4.log
-rw----- 1 root root 999996 Sep 18 16:14 user.5.log
-rw----- 1 root root 999928 Oct 19 09:49 user.6.log
-rw----- 1 root root 999915 Oct 19 17:00 user.7.log
-rw----- 1 root root 999966 Oct 20 16:13 user.8.log
-rw----- 1 root root 999931 Nov 12 09:11 user.9.log
-rw----- 1 root root 864340 Dec 7 15:53 user.log
[AccelinkOLS]
```

3.6.3 命令名称: upload-log server (A.B.C.D) username STRING password STRING

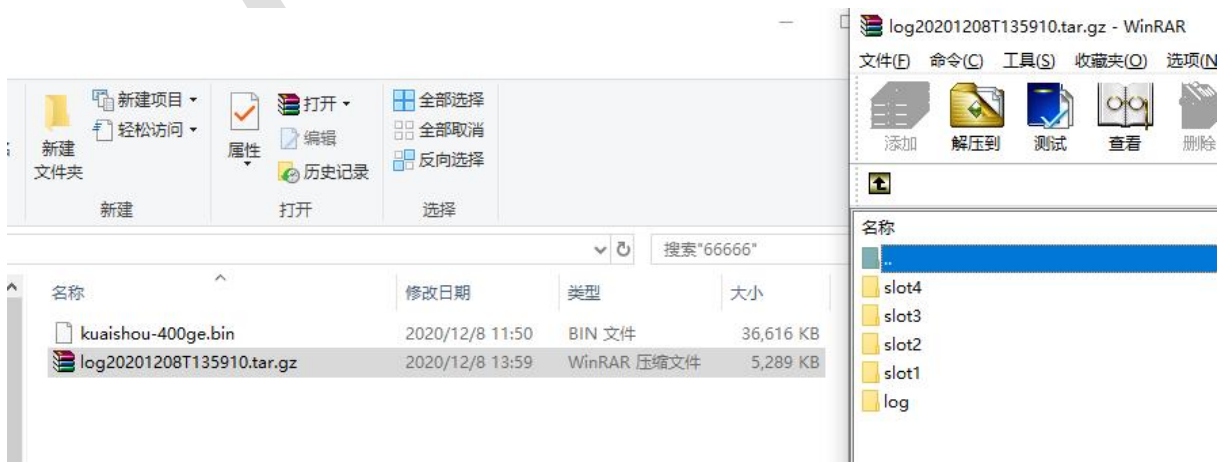
命令描述:

将本地的 log 文件打包提交到指定的远程服务器

参数说明:

注意, 远端的服务器必须工作正常, 而且用户名密码等信息服务预期

```
[Accelink]upload-log server 10.3.22.143 username admin password 123456
File log20201208T135910.tar.gz has been uploaded to 10.3.22.143.
[Accelink]
[Accelink]
[Accelink]
```



3.7 告警管理

3.7.1 命令名称：display alarm

命令描述：

显示当前的告警信息

```
[ACCELINK]display alarm
Date                               Alarm
-----
2022-01-26 00:02:21 [NTP-1                               Name:NTP-SYNC-FAIL                               Level:MAJOR 00001111 NTP-SYNC FAIL 1.2.3.4]
2022-01-26 00:04:29 [TRANSCEIVER-1-1-C2 Name:INPOWER-LOW                               Level:CRITICAL 00021181 INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-1-C2 Name:LANE-INPOWER-LOW                         Level:CRITICAL 00021185 LANE INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-1-C2 Name:R-LOS                                     Level:CRITICAL 0002118C CLIENT R LOS ]
2022-01-26 00:04:27 [TRANSCEIVER-1-1-C2 Name:RX-LOCAL-FLT                             Level:CRITICAL 0002118D RX LOCAL FAULT ]
2022-01-26 00:04:27 [TRANSCEIVER-1-1-C2 Name:TX-LOCAL-FLT                             Level:CRITICAL 0002118F TX LOCAL FAULT ]
2022-01-26 00:04:27 [TRANSCEIVER-1-1-C2 Name:RX-LOS-SYNC                             Level:CRITICAL 0002119A RX LOS SYNC ]
2022-01-26 00:04:29 [TRANSCEIVER-1-1-C3 Name:INPOWER-LOW                               Level:CRITICAL 00031181 INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-1-C3 Name:LANE-INPOWER-LOW                         Level:CRITICAL 00031185 LANE INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-1-C3 Name:R-LOS                                     Level:CRITICAL 0003118C CLIENT R LOS ]
2022-01-26 00:04:27 [TRANSCEIVER-1-1-C3 Name:RX-LOCAL-FLT                             Level:CRITICAL 0003118D RX LOCAL FAULT ]
2022-01-26 00:04:27 [TRANSCEIVER-1-1-C3 Name:TX-LOCAL-FLT                             Level:CRITICAL 0003118F TX LOCAL FAULT ]
2022-01-26 00:04:27 [TRANSCEIVER-1-1-C3 Name:RX-LOS-SYNC                             Level:CRITICAL 0003119A RX LOS SYNC ]
2022-01-26 00:04:29 [TRANSCEIVER-1-1-C4 Name:INPOWER-LOW                               Level:CRITICAL 00041181 INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-1-C4 Name:LANE-INPOWER-LOW                         Level:CRITICAL 00041185 LANE INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-1-C4 Name:R-LOS                                     Level:CRITICAL 0004118C CLIENT R LOS ]
2022-01-26 00:04:27 [TRANSCEIVER-1-1-C4 Name:RX-LOCAL-FLT                             Level:CRITICAL 0004118D RX LOCAL FAULT ]
2022-01-26 00:04:27 [TRANSCEIVER-1-1-C4 Name:TX-LOCAL-FLT                             Level:CRITICAL 0004118F TX LOCAL FAULT ]
2022-01-26 00:04:27 [TRANSCEIVER-1-1-C4 Name:RX-LOS-SYNC                             Level:CRITICAL 0004119A RX LOS SYNC ]
2022-01-26 00:02:21 [NTP-2                               Name:NTP-SYNC-FAIL                               Level:MAJOR 00002111 NTP-SYNC FAIL 2.3.4.5]
2022-01-26 00:04:29 [TRANSCEIVER-1-2-C2 Name:INPOWER-LOW                               Level:CRITICAL 00022181 INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-2-C2 Name:LANE-INPOWER-LOW                         Level:CRITICAL 00022185 LANE INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-2-C2 Name:R-LOS                                     Level:CRITICAL 0002218C CLIENT R LOS ]
2022-01-26 00:04:27 [TRANSCEIVER-1-2-C2 Name:RX-LOCAL-FLT                             Level:CRITICAL 0002218D RX LOCAL FAULT ]
2022-01-26 00:04:27 [TRANSCEIVER-1-2-C2 Name:TX-LOCAL-FLT                             Level:CRITICAL 0002218F TX LOCAL FAULT ]
2022-01-26 00:04:27 [TRANSCEIVER-1-2-C2 Name:RX-LOS-SYNC                             Level:CRITICAL 0002219A RX LOS SYNC ]
2022-01-26 00:04:29 [TRANSCEIVER-1-2-C3 Name:INPOWER-LOW                               Level:CRITICAL 00032181 INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-2-C3 Name:LANE-INPOWER-LOW                         Level:CRITICAL 00032185 LANE INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-2-C3 Name:R-LOS                                     Level:CRITICAL 0003218C CLIENT R LOS ]
2022-01-26 00:04:27 [TRANSCEIVER-1-2-C3 Name:RX-LOCAL-FLT                             Level:CRITICAL 0003218D RX LOCAL FAULT ]
2022-01-26 00:04:27 [TRANSCEIVER-1-2-C3 Name:TX-LOCAL-FLT                             Level:CRITICAL 0003218F TX LOCAL FAULT ]
2022-01-26 00:04:27 [TRANSCEIVER-1-2-C3 Name:RX-LOS-SYNC                             Level:CRITICAL 0003219A RX LOS SYNC ]
2022-01-26 00:04:29 [TRANSCEIVER-1-2-C4 Name:INPOWER-LOW                               Level:CRITICAL 00042181 INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-2-C4 Name:LANE-INPOWER-LOW                         Level:CRITICAL 00042185 LANE INPOWER LOW ]
2022-01-26 00:04:29 [TRANSCEIVER-1-2-C4 Name:R-LOS                                     Level:CRITICAL 0004218C CLIENT R LOS ]
```

3.7.2 命令名称：trigger_alarm component_name STRING alarm_id (<1-1000>|all) alarm_status (on|clear|off)

命令描述：

根据告警组件名称以及告警 ID 选择打开或关闭告警状态

3.8 单盘管理

3.8.1 命令名称：display slot

命令描述：

显示单盘相关的信息

```
[Accelink0LS]display slot
The device has 2 slot online:
Id  SN                Type Status Version Boot Temp Real Send/Recv
1   000aac518598 LPU Normal 0.2/1.3 12  60.6 2726 50455/50453
2   000aac518598 LPU Normal 0.2/1.3 13  59.0 2726 50455/50452
[Accelink0LS]
[Accelink0LS]
```


3.8.2 命令名称：slot <1-8>

命令描述：

执行该命令，进入单盘视图，可以操作与单盘相关的命令。

参数说明：

输入对应的 slot 号

```
[AccelinkOLS]slot 1/1
[AccelinkOLS-LPU-1/1]
[AccelinkOLS-LPU-1/1]
[AccelinkOLS-LPU-1/1]?
 cfp2-dco  cfp2-dco module config
 clear      clear operation
 client     client port
 cls        clear screen
 config     config command
 display    show running system information
 exit       Exit current mode and down to previous mode
 fpga       fpga module config
 list       Print command list
 qsfp28     QSFP28 module config
 quit       Quit current mode and down to previous mode
 reboot     Reboot the device
 reset      Reset optical module
 return     End current mode and down to previous mode.
 vega       vega module config
[AccelinkOLS-LPU-1/1]_
```

3.8.3 命令名称：reset slot <1-8> hard

命令描述：

复位指定的单盘(单盘 CPU 管脚复位，主控通过主板上 fpga 拉单盘复位管脚)

3.8.4 命令名称：reset slot <1-8> cold

命令描述：

复位指定的单盘(单盘掉电复位，主控通过主板上 fpga 拉单盘电源控制芯片管脚)

3.8.5 命令名称：reset slot <1-8> warm

命令描述：

复位指定的单盘(单盘热复位，主控发送一个命令，给单盘，单盘自行重启 MCU)

3.9 单盘业务管理

在 system-view 视图内执行 slot <1-8>可以切换到对应的单盘视图，单盘视图可以对单盘进行对应的管理。

3.9.1 命令名称：reset cfp2-dco

命令描述：

硬复位 cfp2-dco 模块

3.9.2 命令名称：reset client <1-4>

命令描述：

硬复位客户侧指定客户侧模块

3.9.3 命令名称：display cfp2-dco

命令描述：

显示 line 侧的当前状态(线路侧激光器开关状态、频率、输入输出功率等)信息

密级：

```
[AccelinkOLS-LPU-1/1]display cfp2-dco
```

CFP2_DCO Module No.1 information:

Gene Info:

```
Tx Laser Mode      : ON
Power Mode         : Normal Power
Key Status         : Normal
MdDev Mode Value   : 0x00
MdDev KeyStatus Value : 0x00
MdDev Voltage Value : 3364 mV
MdDev Current Value : 6075 mA
MdDev Power Value  : 20436 mW
```

Manu Info:

```
Vendor Name       : Accelink
Vendor PN         : OCT-04C2-M3CN2
Vendor SN         : OCT212854591
Vendor DateCode   : 20210720
```

Basic Info:

```
Connector Type     : oc-opt-types:LC_CONNECTOR
Ethernet App Code   : oc-opt-types:ETH_UNDEFINED
Host Lane Number    : 8
Network Lane Number : 1
Device Technology   : DBR,N/A
Transceiver Temperature Th. : -5.00 C, 80.00 C
VCC Alarm Th.      : 3.13 V, 3.46 V
Tx Power Alarm Th.  : 0.00 dBm, 0.00 dBm
Rx Power Alarm Th.  : -28.00 dBm, 13.00 dBm
Tx Min Laser Freq   : 191275.00 GHZ
Rx Min Laser Freq   : 191275.00 GHZ
Tx Rx Max Laser Freq : 196125.00 GHZ
Laser Tuning Capability : 6.25 GHZ 12.5 GHZ 25 GHZ 50 GHZ 100 GHZ
```

Real Info:

```
Module State       : 0x0020
General Status Latch : 0x0046
Alarm and Warning Latch : 0x0000
Module Temperature  : 49.00 C
Module Power Supply : 3.30 V
SOA Bias           : 0.00 mA
Tx Laser Bias       : 265.00 mA
Network Alm warning 1 Latch : 0x0000
Network Alm warning 2 Latch : 0x0000
TX Channel Ctrl     : 6.25 GHz grid spacing, channel number:777
Network TX Laser Output : 16.02 dBm
Network TX Laser Temp : 45.86 C
Desired TX Output    : 0.00 dBm
TX Current Freq      : 196125.00 GHZ
RX Current Freq      : 196125.00 GHZ
Current Output Power  : 0.00 dBm
Current Input Power   : -1.17 dBm
```

3.9.4 命令名称：display client

命令描述：

显示客户侧模块相关信息（激光器开关状态、输入输出光功率等）

密级：

```
<ACCELINK> system-view
[ACCELINK]
[ACCELINK]s1
[ACCELINK]slot 1/1
[ACCELINK-LPU-1/1]di
[ACCELINK-LPU-1/1]display cl
[ACCELINK-LPU-1/1]display client

    QSFP28 Module No.1 information:
    Tx Laser Mode                : ON
    Power Mode                    : Normal Power
    Key Status                    : Normal
    MdDev Mode Value              : 0x12
    MdDev KeyStatus Value         : 0x00
    MdDev Reserved Value         : 0x06
    ExSpecCompCodes               : 100GBASE-LR4
    Bit Rate                      : 25.8 Gbps
    Wave Length                   : 1310.00 nm
    Temperature Alarm Th.         : -5.00 C, 75.00 C
    VCC Alarm Th.                 : 2.97 V, 3.63 V
    Rx Power Alarm Th.            : -13.60 dBm, 5.50 dBm
    Tx Power Alarm Th.            : -6.30 dBm, 5.50 dBm
    Tx Bias Alarm Th.             : 20.00 mA, 80.00 mA
    Module Temperature             : 42.27 C
    Supply Voltage                 : 3.37 V
    Rx Power                      : 0.12 dBm,-0.87 dBm,1.07 dBm,1.05 dBm
    Tx Power                      : 3.27 dBm,2.03 dBm,2.32 dBm,2.76 dBm
    Tx Bias                       : 40.18 mA,41.26 mA,38.02 mA,41.26 mA
    Vendor Name                   : ACCELINK
    Vendor PN                     : OEM-M1006113Q28X
    Vendor RN                     : 01
    Vendor SN                     : IN203702050044
    Date Code                     : 20091019
```

3.9.5 命令名称：display (client|line) cfg-info

命令描述：

显示(客户侧|线路侧)的配置信息

```
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]display client cfg-info

    Lpu Client configure Information:
    No. Lldp Als Rsfec fMode Loop      Service TtiAuto Ttisend
    1   on   off off   auto  none    100ge    false
    2   on   off off   auto  none    100ge    false
    3   on   off off   auto  none    100ge    false
    4   on   off off   auto  none    100ge    false

[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]display line cfg-info

    Lpu Line configure Information:
    No. Loop      Ntwk-Mode  Ntwk-Cfg  Tti-Auto  Tti-Send
    1   none      enhanced   25        false

[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.6 命令名称：display alarm-threshold line-rx-pow

命令描述：

显示线路侧接收告警门限

密级：

```
[ACCELINK-LPU-1/1]display alarm-threshold line-rx-pow
      slot1 CFP2_DCO line-rx-low alarm-threshold : -40.00
      slot1 CFP2_DCO line-rx-high alarm-threshold : 8.00
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.7 命令名称：display alarm-threshold pre-fec-high

命令描述：

显示线路侧 FEC 告警门限

```
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]display alarm-threshold pre-fec-high
      slot1 pre-fec-high alarm-threshold : 1.000000e-02
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.8 命令名称：display cfp2-dco cd-range

命令描述：

显示客户侧指定 cfp2-dco 模块的色散收敛范围

```
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]display cfp2-dco cd-range
      cd search range : -3400 ~ -200
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.9 命令名称：display cfp2-dco client <1-4> dm-mode

命令描述：

显示客户侧指定 cfp2-dco 模块的时延测量模式信息

```
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]display cfp2-dco client 1 dm-mode
      Dm-Mode : transparent
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.10 命令名称：display cfp2-dco client <1-4> dm-result

命令描述：

显示客户侧指定 cfp2-dco 模块的时延测量结果信息

密级：

```
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]display cfp2-dco client 1 dm-result
Dm-Result                : 0.000 us
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.11 命令名称：display cfp2-dco client <1-4> loopmode

命令描述：

显示客户侧指定端口的环回模式信息

```
[ACCELINK-LPU-1/1]display cfp2-dco client 1 loopmode
loopmode                  : none
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.12 命令名称：display cfp2-dco freqmini

命令描述：

显示 cfp2-dco 模块的最小频率信息

3.9.13 命令名称：display cfp2-dco line dm-mode

命令描述：

显示线路侧 cfp2-dco 模块的时延测量模式信息

3.9.14 命令名称：display cfp2-dco line dm-result

命令描述：

显示线路侧 cfp2-dco 模块的时延测量结果信息

```
[ACCELINK-LPU-1/1]display cfp2-dco line dm-mode
Dm-Mode                   : transparent
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.15 命令名称：display cfp2-dco line loopmode

命令描述：

显示线路侧的环回模式

```
[ACCELINK-LPU-1/1]display cfp2-dco line loopmode
loopmode                : none
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.16 命令名称：display cfp2-dco operation-mode

命令描述：

显示 cfp2-dco 模块当前的 operation-mode 模式

```
[ACCELINK-LPU-1/1]display cfp2-dco operation-mode
NTWK_CFG                : 400G-400ZR-oFEC-16QAM-1-SE
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.17 命令名称：display client <1-4> (tx-laser|als)

命令描述：

显示客户侧指定接口的（发射激光器|als 模式）的信息

```
[ACCELINK-LPU-1/4]display client 1 tx-laser
QSFP28 No.1 tx-laser state: OFF
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]display client 1 als
QSFP28 No.1 als state: ON
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```

3.9.18 命令名称：display client <1-4> lf-debounce

命令描述：

显示客户侧指定接口的 link down delay 信息

```
[ACCELINK-LPU-1/4]display client 1 lf-debounce
lf switch state          : ON
holdoff time(config)     : 1000
holdoff time(real)       : 1000
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```

3.9.19 命令名称：display fpga version

命令描述：

显示单盘上 fpga 的版本信息

```
[ACCELINK-LPU-1/4]display fpga version
version          : 020a
date             : 2021-09-08
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```

3.9.20 命令名称：display lldp <1-4>

命令描述：

显示指定接口的 lldp 信息

参数说明：

无

```
[ACCELINK-LPU-1/1]display lldp 1
Global LLDP information          : ENABLE
Client No.1 LLDP information:
  Ingress LLDP information:
    Chassis ID Type              : MAC address
    Chassis ID                   : 50-0f-80-20-6f-e8
    Port ID Type                 : Interface name
    Port ID                      : Ethernet1/52
    TTL                          : 120 s
    Port Description              : Ethernet1/52
    System Name                  : CISCO_N3K
    System Description            : Cisco Nexus operating System (NX-OS) Software 7.0(3)F3(3)
    2002-2017, cisco Systems, Inc. All rights reserved.
    System Capabilities          :
      Bridge                     : Enabled
      Router                     : Enabled
    Management Addr Type         : all 802
    Management Addr              : 50-0f-80-20-6f-e8
    Interface                    : ifIndex 83886080
    Organizationally Specific Info:
      :
      : PVID: 2
  Egress LLDP information:
    Chassis ID Type              : MAC address
    Chassis ID                   : 50-0f-80-20-6f-e4
    Port ID Type                 : Interface name
    Port ID                      : Ethernet1/51
    TTL                          : 120 s
    Port Description              : Ethernet1/51
    System Name                  : CISCO_N3K
    System Description            : Cisco Nexus operating System (NX-OS) Software 7.0(3)F3(3)
    2002-2017, cisco Systems, Inc. All rights reserved.
    System Capabilities          :
      Bridge                     : Enabled
      Router                     : Enabled
    Management Addr Type         : all 802
    Management Addr              : 50-0f-80-20-6f-e4
    Interface                    : ifIndex 83886080
    Organizationally Specific Info:
      :
      : PVID: 2
```

3.9.21 命令名称：display lpu manufacture

命令描述：

显示业务板卡的生产制造信息(SN/PN)

密级：

```
[ACCELINK-LPU-1/1]display lpu manufacture

DCI Lpu device information:
Vendor Name      : ACCELINK-O2-400M-100X4
SN               : 000AAC15C70A
PD              : 2021-12-14
Name            : T04X1L-103
PN              : T04X1-103

[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.22 命令名称：display operation-mode

命令描述：

显示 line 侧支持的 Operational -mode 信息

```
[AccelinkOLS]slot 1/1
[AccelinkOLS-LPU-1/1]
[AccelinkOLS-LPU-1/1]
[AccelinkOLS-LPU-1/1]
[AccelinkOLS-LPU-1/1]di
[AccelinkOLS-LPU-1/1]display o
[AccelinkOLS-LPU-1/1]display operation-mode

Mode-Id          Operation Mode
9                200G-200ZR-oFEC-QPSK-1-S
13               200G-200ZR-oFEC-16QAM-1-SE
25               400G-400ZR-oFEC-16QAM-1-SE

[AccelinkOLS-LPU-1/1]
```

3.9.23 命令名称：display statistic <1-4>

命令描述：

显示指定接口的统计信息

```
[ACCELINK-LPU-1/1]display statistic 1
```

```
Client No.1 Statistic information:
```

```
Ethernet Channel information:
```

```

Ingress Packets Count      :      6286
Ingress Octets Count       :    1672062
Ingress CRC Errors Count   :          0
Ingress Pkts Undersize     :          0
Ingress Pkts Oversize      :          0
Ingress Pkts Fragment      :          0
Ingress Pkts Jabber        :          0
Ingress Pkts Broadcast     :          0
Ingress Pkts Multicast     :      6286
Ingress Pkts 64 Octets     :          0
Ingress Pkts 65-127 Octets:          0
Ingress Pkts 128-255 Octets:     2096
Ingress Pkts 256-511 Octets:    4190
Ingress Pkts 512-1023 Octets:          0
Ingress Pkts 1024-1518 Octets:          0
Ingress Pkts 1519-Max Octets:          0
Ingress BIP8 Count         :    0.000000e+00
Ingress BIP8 Error Count   :    0.000000e+00
Ingress SH Count           :    0.000000e+00
Ingress SH Error Count     :    0.000000e+00
Ingress PCS Error Count    :    0.000000e+00
Ingress Band Utili         :          0.00

```

```

Egress Packets Count       :      6286
Egress Octets Count        :    1672062
Egress CRC Errors Count    :          0
Egress Pkts Undersize      :          0
Egress Pkts Oversize       :          0
Egress Pkts Fragment       :          0
Egress Pkts Jabber         :          0
Egress Pkts Broadcast      :          0
Egress Pkts Multicast      :      6286
Egress Pkts 64 Octets      :          0
Egress Pkts 65-127 Octets:          0
Egress Pkts 128-255 Octets:     2096
Egress Pkts 256-511 Octets:    4190
Egress Pkts 512-1023 Octets:          0
Egress Pkts 1024-1518 Octets:          0
Egress Pkts 1519-Max Octets:          0
Egress BIP8 Count          :    0.000000e+00

```

3.9.24 命令名称：display vega (general|real)

命令描述：

显示 vega 信息

[ACCELINK-LPU-1/1]display vega real

VEGA Module Realtime Info:

VEGA Module Interface 1:

```

      Info Len          : 253
      Info Uid          : 0x0110
Interface Comm Info:
      Active Status     : true
      Hrx Channel       : 6
      Operational Mode  : Reverse GearBox
      Hmux Enabled      : false
      Hmux Mode         : RGB
      Physical Channel Mask : f0|f0|c0|c0
      Internal Channel Mask : f0|f0|cc|c0
      Hmux Postion      : SWITCH_TO_MASTER
      PCS ES Since Reset : 0
      PCS SES Since Reset : 0
      PCS UAS Since Reset : 0

```

Interface Host Info:

```

      Channel Num       : 2
      Channel SNR       : 25.753, 25.520
      Channel Link Ready : linkup, linkup
      Fpcs Mode         : RS544
      Ctrl Status       : 4c
      Pre Fec Ber       : 0.000000e+00
      Post Fec Ber      : 0.000000e+00
      Host Fpcs Statics : Since Reset PM Interval
      Uncorrected Code words : 0 0
      Corrected Code words  : 4569 0
      Corrected Bits       : 24124320 0
      Idle Errors         : 71075884385 0
      Invalid Sync Headers : 16549663 0
      Invalid Block Types  : 32409355 0
      Symbol Errors       : 4571 0

```

3.9.25 命令名称: display vega die link_status

命令描述:

显示 vega die 的链接状态

[ACCELINK-LPU-1/1]display vega die link_status

Link status for die 0

```

HRX Link Status
Channel      0 1 2 3 4 5 6 7
Signal Detect . . . . Y Y Y Y
DSP Ready    . . . . Y Y Y Y
FW Lock      . . . . . . . .
PLL Lock     . . . . . . . .

```

```

HRX Interrupts
Channel      0 1 2 3 4 5 6 7
PLL Lock Det . . . . . Y .

```

```

LTX Link Status
Channel      0 1 2 3 4 5 6 7
FW Lock      . . . . . . . .
PLL Lock     Y Y Y Y Y Y Y Y

```

```

LTX Interrupts
Channel      0 1 2 3 4 5 6 7
PLL Lock Det . . . . . . . .
FIFOA Empty  . . . . . . . .
FIFOA Full   . . . . . . . .
FIFOB Empty  . . . . . . . .
FIFOB Full   . . . . . . . .

```

```

HTX Link Status
Channel      0 1 2 3 4 5 6 7
FW Lock      . . . . Y Y Y Y
PLL Lock     . . . . Y Y Y Y

```

```

HTX Interrupts
Channel      0 1 2 3 4 5 6 7
PLL Lock Det . . . . Y Y Y Y
FIFOA Empty  . . . . . . . .
FIFOA Full   . . . . . . . .
FIFOB Empty  . . . . . . . .
FIFOB Full   . . . . . . . .

```

```

LRX Link Status
Channel      0 1 2 3 4 5 6 7
Signal Detect . . . . . . . .
DSP Ready    . . . . . . . .
FW Lock      Y Y Y Y Y Y Y Y
PLL Lock     Y Y Y Y Y Y Y Y

```

```

LRX Interrupts
Channel      0 1 2 3 4 5 6 7
PLL Lock Det . . . . . . . Y

```

3.9.26 命令名称：display vega interface <1-4> line fec-switch

命令描述：

显示 vega 指定接口线路侧的 fec 开关信息

```
[ACCELINK-LPU-1/1]display vega interface 1 line fec-switch
get VEGA interface No.1 line fec switch state:  disable!
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.27 命令名称：cfp2-dco tx (enable|disable)

命令描述：

配置线路侧激光器启或关闭

```
[ACCELINK-LPU-1/1]cfp2-dco tx enable
Set CFP_DCO tx enable success !
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
[ACCELINK-LPU-1/1]
```

3.9.28 命令名称：cfp2-dco client <1-4> dm-mode (transparent|loopback|insertion)

命令描述：

配置指定客户侧的时延测量模式

参数说明：

transparent：透传模式

loopback：环回模式

insertion：插入模式

3.9.29 命令名称：cfp2-dco client <1-4> loopmode (none|facility|terminal)

命令描述：

配置指定客户端上的 cfp2-dco 模块的回环模式

参数说明：

none: 取消环回

facility: 客户侧外环

terminal: 客户侧内环

3.9.30 命令名称: cfp2-dco fast-switching-mode (<0-6>|20)

命令描述:

配置 fast-switching-mode 模式(每种模式对应一种 cd-range 范围)

参数说明:

0: 对应 cd-range(-26000~1000), DCO 默认值

1: 对应 cd-range(-3400~0)

2: 对应 cd-range(-3400~-200)

3: 对应 cd-range(-5000~-1000)

4: 对应 cd-range(-5000~0), 百度默认值

5: 对应 cd-range(-7000~-2000)

6: 对应 cd-range(-7000~0)

20: 对应 cd-range(-3600~0), 快手默认值

3.9.31 命令名称: cfp2-dco line dm-mode (transparent|loopback|insertion)

命令描述:

配置 cfp2-dco 模块线路侧时延测量模式

参数说明:

transparent: 透传模式

loopback: 环回模式

insertion: 插入模式

3.9.32 命令名称: cfp2-dco line loopmode (none|facility|terminal)

命令描述:

配置 cfp2-dco 模块线路侧回环模式

参数说明:

密级：

none: 取消环回

facility: 线路侧外环

terminal: 线路侧内环

3.9.33 命令名称: cfp2-dco operation-mode STRING

命令描述:

配置 cfp2-dco 模块的 operation-mode

参数说明:

STRING: 使用数字表示操作模式, 详细说明参考'display operation-mode'命令

```
[ACCELINK-LPU-1/4]display operation-mode

Mode-Id                Operation Mode
7                      200G-FOIC2-oFEC-QPSK-1-S
9                      200G-200ZR-oFEC-QPSK-1-S
12                     200G-FOIC2-oFEC-16QAM-1-SE
13                     200G-200ZR-oFEC-16QAM-1-SE
23                     400G-FOIC4-oFEC-16QAM-1-SE
25                     400G-400ZR-oFEC-16QAM-1-SE

[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]display cfp2-dco operation-mode
NTWK_CFG                : 200G-200ZR-oFEC-QPSK-1-S
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]cfp2-dco operation-mode 25
Set Cfp2Dco operation mode 25 400G-400ZR-oFEC-16QAM-1-SE success!
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]display cfp2-dco operation-mode
NTWK_CFG                : 400G-400ZR-oFEC-16QAM-1-SE
[ACCELINK-LPU-1/4]
```

3.9.34 命令名称: cfp2-dco tx power STRING

命令描述:

配置 cfp2-dco 模块的发射功率

参数说明:

STRING: 功率值, 单位:dBm, 取值范围 -10.0-3.0(精度:0.01)

```
[ACCELINK-LPU-1/4]cfp2-dco tx power 1.0
CFP2_DCO set tx power 1.0 dBm success !
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```


3.9.35 命令名称：cfp2-dco version

命令描述：

查看 cfp2-dco 模块当前版本

```
[ACCELINK-LPU-1/1]cfp2-dco version
CFP2_DCO Module No. version:
Module HWVer           :          65.32
Module A-SWVer          :          66.10.12
Module B-SWVer          :          66.10.12
Module Upgrade Status   :          0x0500
```

3.9.36 命令名称：cfp2-dco wavelength tx STRING

命令描述：

配置 cfp2-dco 的输出波长 (与线路侧频率配置命令 cfp2-dco freq tx <191275000-196125000>等效)

```
[ACCELINK-LPU-1/4]cfp2-dco wavelength tx 12.750
Set CFP2_DCO tx wavelength C12.750 success(191275 GHz ~ 196125 GHz) !
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]cfp2-dco freq tx 191275000
Set CFP2_DCO Tx Freq 191275000MHZ success(191275GHz ~ 196125GHz) !
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```

3.9.37 命令名称：cfp2-dco freq tx <191275000-196125000>

命令描述：

配置 cfp2-dco 的发射频率(与线路侧波长配置命令 cfp2-dco wavelength tx STRING 等效)

```
[ACCELINK-LPU-1/4]cfp2-dco wavelength tx 12.750
Set CFP2_DCO tx wavelength C12.750 success(191275 GHz ~ 196125 GHz) !
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]cfp2-dco freq tx 191275000
Set CFP2_DCO Tx Freq 191275000MHZ success(191275GHz ~ 196125GHz) !
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```

3.9.38 命令名称：clear cfp2-dco (client|line) interface <1-4> statistics

命令描述：

清除 cfp2-dco 客户侧|线路侧的统计数据

参数说明：

client: 客户侧

line: 线路侧

```
[ACCELINK-LPU-1/4]clear cfp2-dco client interface 1 statistics
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]clear cfp2-dco line interface 1 statistics
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```

3.9.39 命令名称: clear statistic (<1-4>|all)

命令描述:

清除指定客户侧以太网统性能统计值

```
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]clear statistic 1
client-1 statistic has been cleared!
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]clear statistic all
client-all statistic has been cleared!
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```

3.9.40 命令名称: config client <1-4> laser (enable|disable)

命令描述:

配置 client 侧 qsfp28 或者 qsfpdd 的激光器开关状态

参数说明:

端口号, 以及对应的状态

```
[ACCELINK-LPU-1/4]config client 1 laser enable
set client module No.1 laser enable success !
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```

3.9.41 命令名称: config client <1-4> als (on|off)

命令描述:

配置 client 侧单向 ALS 开关

参数说明:

对应端口号以及单向 ALS 开关状态

密级：

```
[ACCELINK-LPU-1/4]config client 1 als on
config slot 4 client 1 als on success.
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```

3.9.42 命令名称：config client <1-4> lf (on|off)

命令描述：

配置 client 侧的 link down delay 开关

参数说明：

对应端口号以及 link down delay 开关状态

```
[ACCELINK-LPU-1/4]config client 1 lf debounce 1000
config slot 4 client 1 hold off time of link down delay 1000 success.
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]conf
[ACCELINK-LPU-1/4]config cl
[ACCELINK-LPU-1/4]config client 1 lf
[ACCELINK-LPU-1/4]config client 1 lf on
[ACCELINK-LPU-1/4]config client 1 lf on
config slot 4 client 1 link down delay on success.
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]di
[ACCELINK-LPU-1/4]display client 1 lf-debounce
lf switch state : ON
holdoff time(config) : 1000
holdoff time(real) : 1000
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```

3.9.43 命令名称：config client <1-4> lf debounce <0-3000>

命令描述：

配置 client 侧的 link down delay（注意：需要先配置 client 侧的 link down delay，再配置 client 侧的 link down delay 开关，client 侧的 link down delay 才会生效）

参数说明：

对应的端口号，以及对应的时间，单位毫秒

```

[ACCELINK-LPU-1/4]config client 1 lf debounce 1000
config slot 4 client 1 hold off time of link down delay 1000 success.
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]conf
[ACCELINK-LPU-1/4]config cl
[ACCELINK-LPU-1/4]config client 1 lf
[ACCELINK-LPU-1/4]config client 1 lf on
[ACCELINK-LPU-1/4]config client 1 lf on
config slot 4 client 1 link down delay on success.
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]di
[ACCELINK-LPU-1/4]display client 1 lf-debounce
lf switch state : ON
holdoff time(config) : 1000
holdoff time(real) : 1000
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]

```

3.9.44 命令名称：config lldp global (enable|disable)

命令描述：

配置全局 LLDP(与客户侧 LLDP 开关配合使用。全局 LLDP 开关关闭，则机框所有板卡的 client 侧 LLDP 开关全部关闭；全局 LLDP 开关开启，则需要手动去开启相应板卡的 client 侧 LLDP 开关)

参数说明：

enable: 使能

disable: 去使能

```

[ACCELINK-LPU-1/4]config lldp global enable
Config global LLDP enable success.
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]

```

3.9.45 命令名称：config client <1-4> lldp (enable|disable)

命令描述：

配置 client 侧 LLDP 的开关状态(与全局 LLDP 开关配合使用。)

参数说明：

enable: 使能 LLDP 开关

disable: 关闭 LLDP 开关

密级：

```
[ACCELINK-LPU-1/4]config client 1 lldp disable
Config slot 4 client 1 LLDP disable success.
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]di
[ACCELINK-LPU-1/4]display cl
[ACCELINK-LPU-1/4]display client c
[ACCELINK-LPU-1/4]display client cfg-info
```

Lpu Client configure Information:

No.	Lldp	Als	Rsfec	fMode	Loop	Service	TtiAuto	Ttisend
1	off	off	off	auto	none	100ge	false	
2	on	off	off	auto	none	100ge	false	
3	on	off	off	auto	none	100ge	false	
4	on	off	off	auto	none	100ge	false	

```
[ACCELINK-LPU-1/4]_
```

3.9.46 命令名称：vega interface <1-4> line fec-switch (enable|disable|auto)

命令描述：

配置客户侧 fec 开关

参数说明：

enable：开启 FEC

disable：关闭 FEC

auto：自适应，由单盘自动识别插入客户侧模块类型，自动选择开启/关闭客户侧 FEC

```
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]vega interface 1 line fec-switch enable
Set VEGA interface No.1 line fec switch enable success!
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]vega interface 1 line fec-switch disable
Set VEGA interface No.1 line fec switch disable success!
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]vega interface 1 line fec-switch auto
Set VEGA interface No.1 line fec switch auto success!
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
[ACCELINK-LPU-1/4]
```